

Integrated Analytic Methodology Using Visual Image and Meta-Data for Product Recommendation

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ABSTRACT

Because the big data industry holds a significant position nowadays, Meta-Data has been adopted rapidly in various fields. In this paper, we introduce a model that exploits Meta-Data consisting of numerous production data to extract features using a pretrained deep learning model with image and text information. Thus, we can build a relational model between the production data to realize a recommendation system. Regarding the dataset, we determined that combining both image and text data is better than using one type of data to achieve a more accurate prediction from the relational model. Concerning the condition of two mixed languages (English and Korean), the method produces a satisfactory result. According to the relational model of the two products, we can design a recommendation system that suggests products that would be of interest to consumers.

Key words : Meta-data, Recommender system, Text information, Visual image information

1. Introduction

A recommender system provides proper recommendations based on customer penchant and product features. The ways to build the system are separated into two sections. The first one is Collaborative Filtering (CF), and the second one is Content-based Filtering. The former one assumes that similar tendency goes into similar products preference and guides to products by understanding related behavior patterns of a specific group of people. The latter one is again separated into User-based CF and Item-based CF. The user one produces a recommendation for a person from the others who share similar interests or tendencies. A typical example is Friend Recommendation System that recommends production to a person based on what his or her friends prefer or interested in. Item one is a filtering system that guides a user to check what other people purchase with

similar penchant. Content-based Filtering analyzes the feature of products and makes proper recommendations. For instance, the system interprets the features of the story and characters of a movie or genre and beat of a song, and then produces content that is similar to the user's interests. Recently, Model-based filtering is widely used combine with Collaborative and Content-Based filtering models to analyze user's hidden features and to build adequate models. The goal of this model is analyzing the reason why the users prefer specific production and make a proper recommendation with high accuracy. To understand the hidden features of customers, we need sophisticated algorithms to build a recommender system based on deep learning model and filtering techniques. In this paper, we use image classification and natural language processing that are developed on deep learning algorithms to understand the visual image and integrated Meta-data analysis method for production recommendation. Specifically, we aim to make a system using Meta-data consisting of commodity data and deep learning model to train features of the paired contents or goods in specific relation and to recommend ones that are related to the user's preference.

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In addition, we use Amazon dataset (<http://deepyeti.ucsd.edu/jianmo/amazon/index.html>) to prove the accuracy and usefulness of the integrated model made up of features from production images and Meta-data and introduce a method that performs in an exceptional condition when two languages are mixed during the process using natural language model to extract features from Meta-data.

2. Related Work

The image Classification deep learning model has been improved since AlexNet wins with achieving higher accuracy for image classification algorithm from millions of images in the competition of ImageNet Large Scale Visual Recognition Challenge (ILSVRC) [1]. While the previous models which use narrow architectures, AlexNet uses Convolutional Neural Network with a parallel structure to handle image features and to analyze the increased number of weights due to deepen structure using two parallelly structured GPUs and then obtain outstanding outcome [2]. In addition, VGGNet, GoogLeNet, and ResNet with deep architecture outperform in ILSVRC hold after 2012. VGGNet uses 3×3 size filters for twice times deeper structures as much as AlexNet and proves the depth of architecture is essential for efficiency [3]. GooLeNet uses a 1×1 size filter during the convolution process that makes lesser counts of feature maps that burdening of computing and extracts diverse features [4]. ResNet realizes an unusual deep model makes it worse to classify images, so the model provides a method that connects shortcut that adds input and output up to minimize residuals for training [5]. Thus, ResNet with an architecture approximately seven times deeper than GooLeNet outperforms human for image classification. A deeper convolutional neural network promotes the performance of deep learning image classification but requires more computational capacity. Increased computation challenges can be solved through an operation process device such as a graphic process device, but it remains to be seen as a mobile platform that is hardly equipped with such advanced software. To overcome the issue, MobileNet is based on efficient convolutional computation is created and this model is developed through update from V1 [6], V2 [7] to V3 [8]. In this paper, we use MobileNetV3 to excerpt image features from production in consideration of mobile device expansion and computation efficiency. One-hot Encoding allows the transformation of human language to numeric values that computers use to train. The method builds

word clusters and returns text locations what we want to indicate as 1 and the rest of these are as 0 within a vector which is word set size. However, when we express the words in this way, vector size is increased as the word set does so that causes space waste. So, Word Embedding that manifests fixed real number vectors regardless of the size of word set is introduced, and Recurrent Neural Network (RNN) is used to handle combined sentences that consist of words not only simple word [9]. The recurrent network reuses current data for the next process stage, but it probably causes an issue that the data is not transmitted enough to the end as the sentence gets longer. To address the problem, a system based on Long Short-Term Memory (LSTM) is suggested that removes impractical memory and accepts essential one [10]. After that, Sequence to Sequence (Seq2Seq) architecture is applied to print sentences out through the compression process from input sentences [11]. Nevertheless, data loss occurs during the process, so the Attention technique solves the problem with input data once again as a reference in the output stage [12]. Transformer structure that builds multiple layers not to complement Attention method but to improve sentence process capability better than any past approaches [13]. Based upon the architecture, Bidirectional Encoder Representations from Transformers (BERT) is adopted to allow the pre-trained model to deal with natural language with higher performance [14]. Finally, Sentence-BERT (S-BERT) from BERT method is proposed to remove pointless parts and to implement embedding in consideration on context and meaning of a sentence [15]. In this paper, in terms of proficiency and efficiency, we use S-BERT method to extract text features of a production. It has been studied to build recommend systems from deep learning algorithms in different ways. Especially, the deep learning model is essential for handling visual image and text processes [16]. Research conducts to analyze the relation between purchased and viewed commodities using extracted product image features based on pre-trained image classification without any input information [17].

3. Proposed Method

Fig. 1 explains a whole process to build a predicting model for the relation between two productions from image and text data. Using the two images of the products, pre-trained image classification model MobileNetV3-Large extracts Visual Image Feature Vector. In addition, we produce Text Feature Vector through S-BERT that is a pre-trained language model with text

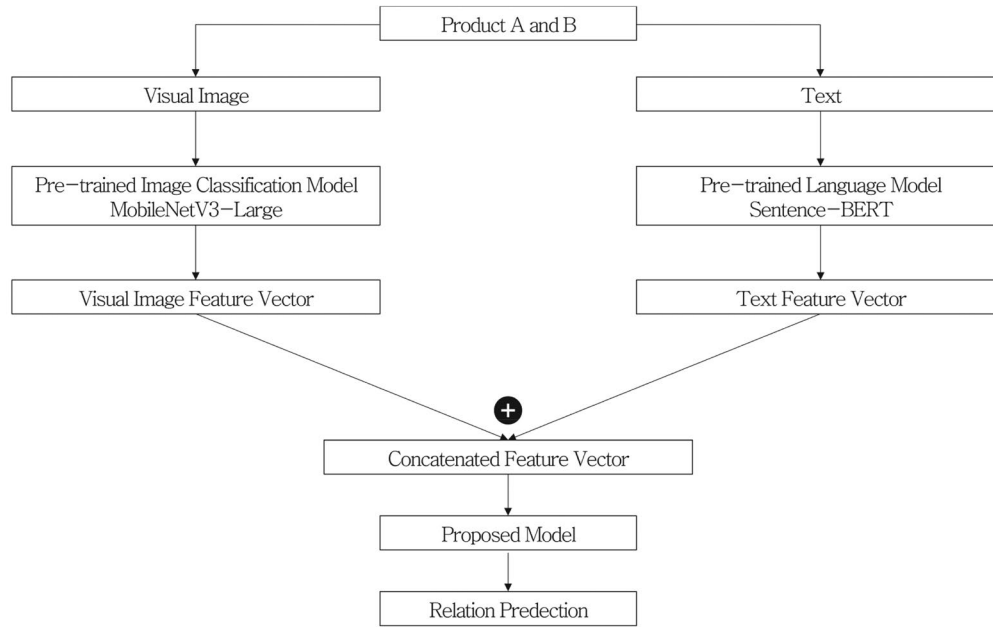


Fig. 1. The whole process to build a predicting model for the relation between two productions.

data from the productions. We create Concatenated Feature Vector by connecting image and text information then use it to train the model that predicts the relation of those products.

3.1 Visual image feature

Table 1 indicates that a process extracts visual image features through MobileNetV3-Large. We use a commodity image with 224 vertical and horizontal dimensions as input data to the model (MobileNetV3-Large) which is already trained and extract 960 size vectors through the image handling process to use production image feature vectors in advance of entering classification layers.

The image size is generally represented by Height (H) × Width (W) × Channels (C). The deep learning model receives the image as an input and extracts features while reducing the size of the image through convolution and pooling operations. Fig. 2 shows a simple example of convolution and average pooling operations. The convolution operation performs element-wise product as the filter moves from left to right and from top to bottom by stride, and the sum of all elements becomes output. When processing an image consisting of three channels, the size and number of output channels vary depending on the size, element, and stride of the filter. In the average pooling operation, the filter moves like the one of convolution, and the average value of elements overlapping the filter becomes output. The pooling operations are generally used to summarize the extracted

Table 1. The process to extract visual image features

Input	Operator	Expansion Size	Output Channel
$224 \times 224 \times 3$	Convolution	—	16
$112 \times 112 \times 16$	Bneck	16	16
$112 \times 112 \times 16$	Bneck	64	24
...	...	...	...
$7 \times 7 \times 160$	Bneck	960	160
$7 \times 7 \times 160$	Convolution	—	160
$7 \times 7 \times 960$	Average Pooling	—	960

features.

The Bneck operation is developed by MobileNetV2 [8] and enhanced by MobileNetV3 [9]. It processes images in a similar way to convolution, but it uses expansion methods to reduce memory usage to increase efficiency, reduce image size, and maintain critical information without losing it.

3.2 Text feature

Table 2 demonstrates a process of extracting production text features through pre-trained S-BERT. We use the text data as input to S-BERT and the output vector as commodity text feature vector. We consider two cases that are monolingual (English) and multilingual (English and Korean) sets.

When it comes to mono-language (English) sentences, all-

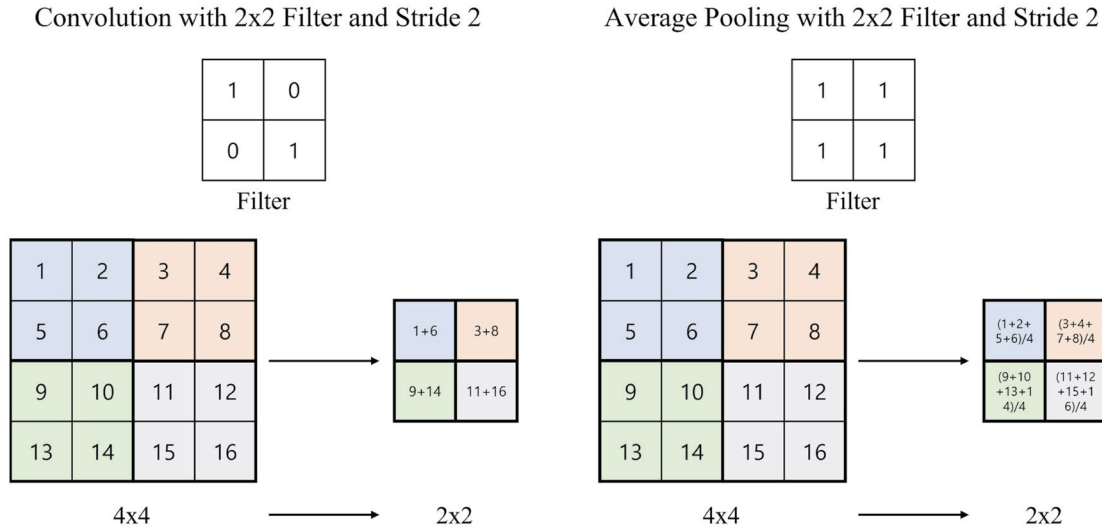


Fig. 2. The simple examples of convolution and average pooling operations.

Table 2. The process to extract production text features

Monolingual Text (English)		Multilingual Text (English and Korean)	
Pre-trained S-BERT Model (English)	Pre-trained S-BERT Model (Single)	Pre-trained S-BERT Model (Separate)	
		Pre-trained S-BERT Model (English)	Pre-trained S-BERT Model (Korean)
Text Feature Vector [1×384]	Text Feature Vector [1×384]	English Text Feature Vector [1×384]	Korean Text Feature Vector [1×786]
		Concatenated Text Feature Vector [1×1152]	

Table 3. The process to produce Concatenated Feature Vector

	Product A		Product B	
Input	Visual Image	Text	Visual Image	Text
Pre-trained model	MobileNetV3	S-BERT	MobileNetV3	S-BERT
Feature vector	u_i	u_t	v_i	v_t
Concatenated Feature Vector		$V = [u_i \ v_i \ u_i - v_i \ u_t \ v_t \ u_t - v_t]$		

MiniLM-L6-v2 (https://www.sbert.net/docs/pretrained_models.html) that is designed to embed the language into vectors which is used to gain 384 size text feature vectors. For the multi-language sentences (English and Korean), we have two options to obtain text vectors. First one obtains 384 size feature vectors using praphrase-multilingual-MiniLM-L12-v2 that is trained to embed different language sentences into each one. Another one separates the languages and uses KR-SBERT-V40K-klueNLI-augSTS (<https://github.com/snunlp/KR-SBERT>) to gain 384 and 768 size vectors and connect these materials to get final

1152 size text feature vectors.

3.3 Proposed model

Table 3 illustrates a process to produce Concatenated Feature Vector that is built to train a predicting model of the relation of two productions using the visual image and text data followed by extracting feature vectors via the method we introduce previously. To begin with, we get the image and text feature data from products A and B and then make final feature vectors through the process that connects to data in the order image

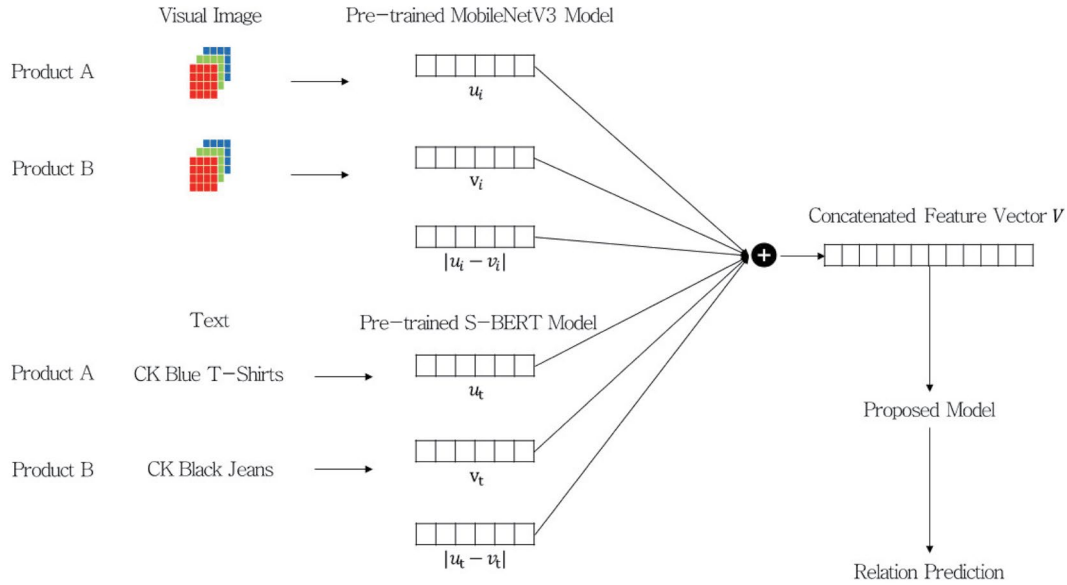


Fig. 3. The process of extracting visual image and text feature vectors from two productions to train the proposed model.

feature vectors from A and B, the vectors in taking the absolute value of the outcome of difference between image vector of A and B, the text feature vector of A and B, the vectors taking the absolute value of the outcome of difference between text vector of A and B. Fig. 3 explicates the input process that extracts specific vectors using the image and text data from two productions for training prediction model that contains the relation of the products. Table 4 exhibits the vector sizes using the final model we mentioned earlier. Each monolingual English and Korean generates 4032 and multilingual Korean and English model creates 4032 as well. In this case, a separated multilingual one gets 6336 sizes.

Deep learning models typically consist of multiple layers for high performance, which pass through the activation function when transferring the output from the previous layer to the next layer. The Rectified Linear Unit (ReLU) and Sigmoid function used in the proposed model are representative activation functions. The ReLU function is the same as (1), and when a negative number is entered, 0 is output, and when a positive number is entered, the input value is returned as it is. The Sigmoid function is the same as (2), and if an input value increases, it becomes close to 1, and if an input value decreases, it becomes close to 0. These characteristics can be used for classification tasks. For example, if the threshold is set to 0.5, and the output value is 0.5 or more, it can be determined to be True, and if it is less than 0.5, it can be determined to be False.

Table 4. The vector sizes for the final model

Text Type and S-BERT Model	Concatenated Feature Vector Size
Monolingual (English)	1×4032
Multilingual (English and Korean) (Single)	1×4032
Multilingual (English and Korean) (Separate)	1×6336

Table 5. The model structure that predicts the relation of the productions

Layer	Input	Sequence	Output
Layer1	V	Linear(V, 1000) ReLU()	1×1000
Layer2	1×1000	Linear(1000, 1)	1×1
Sigmoid	1×1	Sigmoid()	Relation Probability

$$ReLU(x) = \max(0, x) \quad (1)$$

$$Sigmoid(x) = \frac{1}{1 + e^{-x}} \quad (2)$$

Table 5 illustrates a model structure that predicts the relation of the commodities from the final feature vector input. The first layer accepts the feature vectors and reduces them to 1000 sizes and then execute the process through the ReLU function. The second one scales down that to 1 size with the same process. In the final step, the data is sent to Sigmoid function to determine the binary classification of production relation.

Product ID	Relation (Also View)	Relation (Also Buy)	High Resolution Product Image	Title
B00062NHH 0	B015UQ2F3I B01EC8FALE B01F74QQKG	B015FEUANW B015FEUAZ0 B015FEUAF0		Calvin Klein Men's 3-Pack Classic V-Neck T-Shirt
B00063VWS A	B00YROTNNK B00YRZKN4E B03SOUHVK	—		MG Pleated Cotton Twill Hat Band
B0006J31WY	—	B06KENFAOX B02IENFPQA B01PQANVAQ		CliC Adjustable Front Connect Universal Sport Goggle, Black Frame
B000P0XOW Q	—	—		Charades Women's Deluxe Beer Garden Girl Costume Dress with Petticoat

Fig. 4. The part of the monolingual dataset.

Product ID	Relation (Also View)	Relation (Also Buy)	High Resolution Product Image	Title
B00062NHH 0	B015UQ2F3I B01EC8FALE B01F74QQKG	B015FEUANW B015FEUAZ0 B015FEUAF0		Calvin Klein 남성 3팩 클래식 브이넥 T-Shirt
B00063VWS A	B00YROTNNK B00YRZKN4E B03SOUHVK	—		MG Pleated 코튼 트윌 Hat Band
B0006J31WY	—	B06KENFAOX B02IENFPQA B01PQANVAQ		CliC 조정식 전면 연결 범용 클래식 고글, Black Frame
B000P0XOW Q	—	—		Charades 여성 디럭스 비어 가든 Costume Dress with Petticoat

Fig. 5. The part of the multilingual dataset.

4. Experiments

4.1 Datasets

We extract visual image features based on the image classification model that is introduced earlier and use the part of Amazon review data that is exploited to predict purchased produc-

tion relation using images without other inputs. We refine Amazon Fashion Meta-data consists of categorical 186,637 data and reorganize a dataset for the test with the introduced method above. We use ID to differentiate productions, visual images of products for image information, name category for text data, and relation information for related products respectively.

Table 6. The dataset for the experiment

Also View Dataset [2000]		Also Buy Dataset [2000]	
Relation Pairs [1000]	Not Related Pairs [1000]	Relation Pairs [1000]	Not Related Pairs [1000]
Train [70%], Test [30%]		Train [70%], Test [30%]	

Table 7. The hyperparameters for the model

Batch Size	Train Epoch	Learning Rate	Optimizer	Loss Function
20	200	0.001	Adam	Binary Cross Entropy Loss

Table 8. The result for the experiment from the monolingual dataset

Relation	Concatenated Feature Vector V	Test Accuracy (%)
Also View	Visual Image Feature	70.89
	Text Feature	96.60
	MaxAbs Scaled Visual Image Feature + Text Feature	78.50
	MinMax Scaled Visual Image Feature + Text Feature	97.52
	Visual Image Feature + Text Feature	98.15
Also Buy	Visual Image Feature	66.86
	Text Feature	94.62
	MaxAbs Scaled Visual Image Feature + Text Feature	86.49
	MinMax Scaled Visual Image Feature + Text Feature	95.74
	Visual Image Feature + Text Feature	97.20

4.1.1 Monolingual dataset

The part of a monolingual name dataset (English) using reconstructed Amazon Fashion Meta-data is listed in Fig. 4. We understand Also View, and Also Buy relation of the commodities and these materials have image and text data in terms of name category. From the structure, we get the pair of in the relation in Also View or Also Buy and vice versa.

4.1.2 Multilingual dataset

Fig. 5 exhibits a part of production name data with multi-languages (Korean and English) from redefined Amazon Fashion Meta-Data. Each product has the relation of Also View, and Also Buy to other productions and names of image and text data. From the dataset, we get the pair of the relation in Also View or Also Buy and vice versa.

4.2 Results

Table 6 represents the dataset for the suggestion model

experiment. We create randomly 2000 pairs from each 1000 pair of products within Also View relation and not in the one and then divide them into 70% train and 30% test set. In addition, we make Also Buy the same way. Table 7 shows hyperparameters for the model. The following lists are the parameters in the model. Batch size (20), Train Epoch (200), Learning Rate (0.001), Optimizer (Adam), and Loss Function is Binary Cross Entropy.

The result tables articulate an experiment result to compare test accuracies depending on ways to produce final feature vectors. The following lists are indicating the methods. 1. Using only Visual Image Feature. 2. Only Text Feature. 3. Using MaxAbs Scaled Visual Image Feature + Text Feature. 4. Min-Max Scaled Visual Image Feature + Text Feature. 5. Visual Image Feature + Text feature.

4.2.1 Monolingual experiment result

Table 8 demonstrates a result of the experiment from product

Table 9. The result of the experiment from the multilingual dataset

Relation	S-BERT Model	Concatenated Feature Vector V	Test Accuracy (%)
Also View	Multilingual (Single)	Visual Image Feature	70.44
		Text Feature	83.82
		MaxAbs Scaled Visual Image Feature + Text Feature	77.37
		MinMax Scaled Visual Image Feature + Text Feature	77.60
		Visual Image Feature + Text Feature	84.80
	Separate (English, Korean)	Visual Image Feature	70.44
		Text Feature	88.11
		MaxAbs Scaled Visual Image Feature + Text Feature	77.43
		MinMax Scaled Visual Image Feature + Text Feature	81.77
		Visual Image Feature + Text Feature	89.81
Also Buy	Multilingual (Single)	Visual Image Feature	65.58
		Text Feature	81.59
		MaxAbs Scaled Visual Image Feature + Text Feature	74.32
		MinMax Scaled Visual Image Feature + Text Feature	77.20
		Visual Image Feature + Text Feature	82.45
	Separate (English, Korean)	Visual Image Feature	65.58
		Text Feature	82.81
		MaxAbs Scaled Visual Image Feature + Text Feature	67.89
		MinMax Scaled Visual Image Feature + Text Feature	70.78
		Visual Image Feature + Text Feature	84.89

names in the monolingual set (English). We obtain the highest accuracy using Visual Image Feature Vector and Text Feature Vector.

4.2.2 Multilingual experiment result

Table 9 explains a result of the experiment from product names in a multilingual set (English and Korean). We obtain the highest accuracy using Visual Image Feature Vector and Text Feature Vector. In addition, during the process that extracts the Text Vector, we gain the higher accuracy using a separate model than the single one.

5. Conclusion and Discussion

In the Amazon Fashion Meta-data experiment, we understand that it is possible to build a model that predicts the relation of two productions using a text feature vector with the pre-trained deep-learning model.

Especially, using both the visual data and text data achieves a satisfying result, and 80% accuracy is obtained even in the case of multilingual data. We assume that it is feasible to provide a recommendation service of products that are highly related to

ones that users are interested in. However, it remains that further research is needed to understand if the model that we introduce in this paper would work properly on real Meta dataset not only on Amazon one. This is on account of the difference of production types and relations on visual image and text data between a real and the experiment one. We plan to conduct research to devise an environment to collect product data in real and to obtain a normal method to build a model that forecasts relations using image and text data from the productions.

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References

1. Russakovsky O, Deng J, Su H, Krause J, Satheesh S, Ma S, et al. Imagenet large scale visual recognition challenge. Int J Comput

- Vision 2015;115:211-252.
2. Krizhevsky A, Sutskever I, Hinton GE. Imagenet classification with deep convolutional neural networks. *Adv Neur In*: 25; 1106-1114. Nevada; Dec. 2012.
 3. Simonyan K, Zisserman A. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*. 2014.
 4. Szegedy C, Liu W, Jia Y, Sermanet P, Reed S, Anguelov D, et al. Going deeper with convolutions. *Proc Cvpr Ieee*; 1-9. Boston; Jun. 2015.
 5. He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. *Proc Cvpr Ieee*; 770-778. Las Vegas; Jun. 2016.
 6. Howard AG, Zhu M, Chen B, Kalenichenko D, Wang W, Weyand T, et al. Mobilenets: Efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv:1704.04861*. 2017.
 7. Sandler M, Howard A, Zhu M, Zhmoginov A, Chen LC. Mobilenetv2: Inverted residuals and linear bottlenecks. *Proc Cvpr Ieee*; 4510-4520. Salt Lake City; Jun. 2018.
 8. Howard A, Sandler M, Chen B, Wang W, Chen LC, Tan M, et al. Searching for mobilenetv3. *Ieee I Conf Comp Vis*; 1314-1324. Seoul; Nov. 2019.
 9. Mikolov T, Karafiat M, Burget L, Cernocky J, Khudanpur S. Recurrent neural network based language model. *Conf. of the International Speech Communication Association (INTER SPEECH)*; 1045-1048. Makuhari; Sep. 2010.
 10. Sak H, Senior A, Beaufays F. Long short-term memory based recurrent neural network architectures for large vocabulary speech recognition. *arXiv preprint arXiv:1402.1128*. 2014.
 11. Sutskever I, Vinyals O, Le QV. Sequence to sequence learning with neural networks. *Adv Neur In*: 27; 3104-3112. Montreal; Dec. 2014.
 12. Bahdanau D, Cho K, Bengio Y. Neural machine translation by jointly learning to align and translate. *arXiv preprint arXiv:1409.0473*. 2014.
 13. Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, et al. Attention is all you need. *Adv Neur In*: 30; 5998-6008. Long Beach; Dec. 2017.
 14. Devlin J, Chang MW, Lee K, Toutanova K. BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*. 2018.
 15. Reimers N, Gurevych I. Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084*. 2019.
 16. Ni J, Li J, McAuley J. Justifying recommendations using distantly-labeled reviews and fine-grained aspects. *Conf. on Empirical Methods in Natural Language Processing (EMNLP)*; 188-197. Hong Kong; Nov. 2019.
 17. McAuley J, Targett C, Shi Q, Van Den Hengel A. Image-based recommendations on styles and substitutes. *Sigir Forum*; 43-52. Santiago; Aug. 2015.